GradeWise Service

Automated Exam Evaluation using AI

Question Paper > Model Answer > Student Sheet > 4 Results

Start New Evaluation

Evaluation Result

Student ID: MH005

F

19.0 / 40.0

Scoreboard

Question	Score	Max	%	Status
Section A				
Q1	1	1	100% 1/1	Scored
Q2	1	1	100% 1/1	Scored
Q3	1	1	100% 1/1	Scored
Q4	1	1	100% 1/1	Scored
Q5	1	1	100% 1/1	Scored
Q6	0	1	0% 0/1	Not Scored
Q7	1	1	100% 1/1	Scored

Question	Score	Max	%		Status
Q8	0	1	0%	0/1	Not Scored
Section B					
Q9	1	2	50%	1/2	Scored
Q10	0.5	2	25%	0.5/2	Scored
Q11	1	2	50%	1/2	Scored
Q12	1	2	50%	1/2	Scored
Q13	1	2	50%	1/2	Scored
Q14i	0	1	0%	0/1	Not Scored
Q14ii	0	1	0%	0/1	Not Scored
Section C					
Q15	1.5	3	50%	1.5/3	Scored
Q16i	0	1.5	0%	0/1.5	Not Scored
Q16ii	0	1.5	0%	0/1.5	Not Scored
Q17	1.5	3	50%	1.5/3	Scored
Q18	1.5	3	50%	1.5/3	Scored
Section D					

Question	Score	Max	%		Status	
Q19	4	8	50%	4/8	Scored	
Total	19	40	48%		—	—

Question-wise Details

Question 1

1 / 1

The student's answer matches the model answer exactly in terms of the correct option and value. The number of subsets of a set with 3 elements is 2^3 = 8, which corresponds to option (b).

Suggestions

- Ensure to include the option letter and value as in the model answer for clarity.

Question 2

1 / 1

The student's answer matches the model answer exactly in meaning and content, despite minor differences in formatting and capitalization. The student correctly identified option (b) and provided the correct description of the universal set.

Suggestions

- Ensure consistent capitalization for professionalism, e.g., 'Contains all elements under consideration' instead of 'Contains all elements under Consideration'.
- Include the option letter in the same format as the question (e.g., '(b)') for clarity.

Question 3

1 / 1

The student's answer matches the model answer exactly in terms of the correct option and value. Both indicate that cos A is 4/5, which is the correct value given sin A = 3/5 for an acute angle A.

Suggestions

- Ensure to include the reasoning or calculation steps to demonstrate understanding, such as using the Pythagorean identity: cos A = sqrt(1 - sin^2 A) = sqrt(1 - (3/5)^2) = 4/5.
- Write the answer in the exact format requested, but minor differences like 'a)' instead of '(a)' are acceptable.

Question 4

1 / 1

The student's answer matches the model answer exactly in meaning and choice, correctly identifying (c) Many–one relation as the function. Minor differences in formatting and spacing do not affect correctness.

Suggestions

- Maintain consistent formatting with the question options, such as using parentheses around the option letter.
- Avoid spaces around hyphens in compound terms like 'Many–one' to match standard notation.

Question 5

1 / 1

The student's answer matches the model answer exactly in terms of the option chosen and the value of sin 30°, which is 1/2. The slight difference in formatting (parentheses placement) does not affect correctness.

Suggestions

- Maintain consistent formatting with the question options, such as including parentheses around the option letter, to improve clarity and presentation.

Question 6

0 / 1

The correct number of ways to arrange 3 books out of 5 different books is given by the permutation ${}^5P_3 = 5 \times 4 \times 3 = 60$, which corresponds to option (c). The student selected option (a) 10, which is incorrect.

Suggestions

- Review the concept of permutations and how to calculate the number of arrangements when order matters.
- Remember that arranging 3 books out of 5 is a permutation problem, not a combination problem.
- Practice calculating permutations using the formula nPr = n × (n-1) × ... × (n-r+1).

Question 7

1 / 1

The student's answer matches the model answer exactly in meaning and notation, correctly identifying that if A and B are disjoint sets, then their intersection is the empty set ($A \cap B = \varnothing$).

Suggestions

- Ensure to include the parentheses around the option letter as in the model answer for perfect formatting.
- Maintain consistency in notation and formatting to match the model answer exactly.

Question 8

0 / 1

The student's answer 'a) Reflexive' does not match the model answer '(c) Transitive'. The relation 'less than' on natural numbers is transitive, not reflexive. Therefore, the answer is incorrect and no marks can be awarded.

Suggestions

- Review the definitions of reflexive, symmetric, transitive, and equivalence relations.
- Understand that 'less than' is not reflexive because no number is less than itself.
- Recognize that 'less than' is transitive because if $a < b$ and $b < c$, then $a < c$.

Question 9

1 / 2

The student's answer correctly identifies that a set is a collection of objects and mentions that it is written in braces, which aligns with the example part of the model answer. However, the definition is incomplete as it does not specify that the collection must be well-defined. Additionally, the student did not provide an explicit example as requested. Therefore, half marks are awarded for the correct but incomplete definition and partial reference to notation.

Suggestions

- Include the phrase 'well-defined' to clarify that the collection must be clearly specified.
- Provide a concrete example of a set, such as A = {1, 2, 3}, to fully answer the question.
- Write the definition in a complete sentence for clarity.

Question 10

0.5 / 2

The student correctly identified the domain as all real numbers, which is accurate. However, the range is incorrectly stated as all real numbers, whereas the correct range is all real numbers greater than or equal to zero ($x \geq 0$). Since the domain is correct but the range is incorrect, partial credit is awarded.

Suggestions

- Specify the range more precisely as all real numbers greater than or equal to zero.
- Use set notation or interval notation to clearly express domain and range.
- Review the concept that squaring any real number results in a non-negative value, which restricts the range.

Question 11

1 / 2

The student provided the correct final answer (1/2) but did not show any intermediate steps or working. According to the grading rules, this warrants half marks.

Suggestions

- Show the intermediate steps, such as expressing sin 45° and cos 45° as $1/\sqrt{2}$ before multiplying.

- Use consistent notation and spacing, e.g., write 'sin 45° × cos 45°' for clarity.

Question 12

1 / 2

The student provided the correct final numerical value of 20 for the permutation calculation, which matches the model answer's result. However, the student wrote '5P2' instead of '${}^5P_4$' and did not show any working or steps to justify the answer. According to grading rules, correct final answer without steps earns half marks.

Suggestions

- Show the calculation steps explicitly, such as 5 × 4 × 3 × 2, to demonstrate understanding.
- Use the correct notation for the permutation problem (${}^5P_4$) to avoid confusion.
- Clarify how the value 20 was obtained to receive full marks.

Question 13

1 / 2

The student provided the correct final answer (16) but did not show the calculation or reasoning steps (2^4 = 16). According to the grading rules, this warrants half marks.

Suggestions

- Include the calculation or reasoning steps to show how the answer was derived, e.g., 'Number of elements in power set = 2^4 = 16.'
- Use consistent notation and explain the formula used for clarity.

Question 14i

0 / 1

No student answer available.

Suggestions

- Provide an answer to receive credit.

Question 14ii

0 / 1

No student answer available.

Suggestions

- Provide an answer to receive credit.

Question 15

1.5 / 3

The student correctly states the left-hand side (LHS) and the right-hand side (RHS) of the identity and concludes that the RHS equals 2 cosecθ. However, the student does not show any intermediate steps or algebraic manipulations to prove the identity. Since the final answer is correct but lacks working, half marks are awarded.

Suggestions

- Show the algebraic steps to combine the two fractions on the LHS.
- Demonstrate the expansion and simplification of the numerator and denominator.
- Explicitly show how the expression reduces to 2 cosecθ to fully prove the identity.

Question 16i

0 / 1.5

No student answer available.

Suggestions

- Provide an answer to receive credit.

Question 16ii

0 / 1.5

No student answer available.

Suggestions

- Provide an answer to receive credit.

Question 17

1.5 / 3

The student provided the correct final answer (60) and correctly identified the permutation notation 5P3. However, the student did not show the calculation steps ($5 \times 4 \times 3$) as in the model answer. According to the grading rules, correct final answer without steps earns half marks.

Suggestions

- Include the calculation steps to show how the final answer is derived (e.g., $5 \times 4 \times 3$).
- Use consistent notation and formatting to improve clarity.

Question 18

1.5 / 3

The student correctly identified that the relation is reflexive and symmetric, which matches part of the model answer. However, the student did not mention whether the relation is transitive, which is an important part of the question and the model answer. Since the final conclusion about transitivity and the equivalence relation status is missing, the answer is incomplete.

Suggestions

- Include a statement about whether the relation is transitive or not.
- Explain briefly why the relation is transitive to demonstrate understanding.
- Conclude whether the relation is an equivalence relation based on reflexivity, symmetry, and transitivity.

Question 19

4 / 8

The student correctly identified the key trigonometric identity $1 + \tan^2\theta = \sec^2\theta$ and used it to equate the left-hand side (LHS) to the right-hand side (RHS). However, the student did not explicitly show the substitution step or the multiplication with $(1 + \sin^2\theta)$, nor did they clearly write out the intermediate expression $\sec^2\theta(1 + \sin^2\theta)$. The answer is correct but lacks the complete step-by-step working as shown in the model answer.

Suggestions

- Show the substitution explicitly: replace $(1 + \tan^2\theta)$ with $\sec^2\theta$ in the expression.

- Write out the intermediate step: $(1 + \tan^2\theta)(1 + \sin^2\theta) = \sec^2\theta(1 + \sin^2\theta)$.
- Conclude clearly that LHS equals RHS after substitution to complete the proof.